LEARN / DATA CENTERS / BEGINNER EDITION

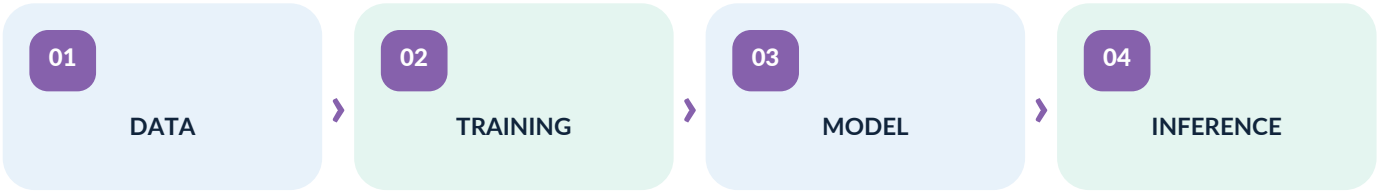
AI (Artificial Intelligence) · KV (Key-Value)

HOW AI CLUSTERS WORK

Training changes model parameters; inference uses a trained model to answer.

SCAN / LEARN

THE SYSTEM AT A GLANCE



FOUR IDEAS THAT MATTER

1 TRAINING

Repeatedly process examples and adjust a model's internal parameters. May use many graphics processors together.

2 CHECKPOINT

Saved training state allows work to resume after interruption. It takes time and storage to save.

3 INFERENCE

Use an existing model to process new input. The serving system manages incoming requests and resources.

4 KV CACHE (KEY-VALUE CACHE)

During some transformer-model inference, stored intermediate attention data helps avoid recomputing earlier tokens.

MEMORY HOOK

Training builds the recipe. Inference uses it. Both need data centers.

DISCUSS WITHOUT COMPUTERS

If the model is already trained, why might a response still be slow?

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